

Formalisation of Constructible Polygons

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Sections

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Background

- Constructible numbers are complex numbers “obtainable” using compass and straight edge.
- Goes back to the Ancient Greeks with problems like trisecting an angle and doubling a cube.
- Similarly, finding which regular n -gons are constructible is another problem that the Ancient Greeks considered.
- Although geometric problem, can be reduced to algebraic problem involving Galois theory.

Rigorous Definitions

We take complex numbers to be points in the complex plane $\mathbb{C}$. To rigorously define what it means to be constructible, we define the operations possible with straightedge and compass:

Definition (Constructions)

C1. From two distinct points $\alpha \neq \beta$, we can draw the line l that goes through these two points.

C2. From two distinct points $\alpha \neq \beta$ and a third point γ , we can draw the circle C with centre γ whose radius is $|\alpha - \beta|$.

Rigorous Definitions

Definition (Points)

We get new constructible numbers in the following ways:

- P1. The intersection of two distinct lines l_1 and l_2 gives a new point.
- P2. The intersection of a line l and a circle C gives a new point.
- P3. The intersection of distinct circles C_1 and C_2 gives a new point.

Definition

A complex number α is constructable if there exists a finite sequence of constructions C_1, C_2, P_1, P_2 and P_3 that begins with 0 and 1 and ends with α .

The Field of Constructable Numbers

Theorem (1)

The set $C = \{\alpha \in \mathbb{C} : \alpha \text{ is constructable}\}$ is a subfield of $\mathbb{C}$.

Furthermore:

- *Let $\alpha = a + ib \in \mathbb{C}$, where $a, b \in \mathbb{R}$. Then $\alpha \in C$ if and only if $a, b \in C$.*
- *$\alpha \in C$ implies that $\sqrt{\alpha} \in C$.*

Important Theorems

Theorem (2)

Let α be a complex number. Then $\alpha \in \mathcal{C}$ if and only if there exist subfields

$$\mathbb{Q} = F_0 \subset F_1 \subset \cdots \subset F_{n-1} \subset F_n \subset \mathbb{C}$$

such that $\alpha \in F_n$ and $[F_i : F_{i-1}] = 2$

In characteristic 0 fields, we have that $[F_i : F_{i-1}] = 2$ implies that $F_i = F_{i-1}(\sqrt{\beta})$ for some $\beta \in F_{i-1}$, which characterizes the constructible numbers as those that can be formed starting from 0 and 1 and only using addition, subtraction, division, multiplication, and taking square roots.

Important Theorems

We also see that this implies the following corollary:

Theorem (Corollary)

If $\alpha \in C$, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^m$ for some $m \geq 0$.

Important Theorems

A partial converse to the previous corollary gives a much more workable method for determining when a complex number is constructible:

Theorem (3)

Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$ and let $\mathbb{Q} \subset L$ be a splitting field of the minimal polynomial of α over $\mathbb{Q}$. Then $\alpha \in C$ if and only if $[L : \mathbb{Q}]$ is a power of two.

Implementation in Lean

- Most of the work in formalizing constructible numbers was already implemented in Lean, thanks to work by fellow Bonn student Ludwig Monnerjahn [Mon24]
- The library defines lines, circles, and the constructions C1, C2, P1, P2, P3.
- The field C is defined as

$$C = M_{\infty} = \bigcup_{n \in \mathbb{N}} M_i$$

where $M_0 = \{0, 1\}$ and M_{i+1} is the set defined by the union of M_i and all points that can be constructed via the intersection of lines and circles directly obtainable from points in M_i .

Implementation in Lean

- The library then proves various properties about lines and circles and that $\mathcal{C}$ is a subfield of $\mathbb{C}$.
- Also proves all the important theorems mentioned except for the last theorem.

Constructable Polygons

Definition

A constructable polygon is a polygon which can be constructed using a straight edge and compass.

Theorem (Gauss-Wantzel Theorem)

Let $n \geq 3$. The regular n -gon is constructable if and only if

$$n = 2^s p_1 \cdots p_r$$

where $s \geq 0$ is an integer and the p_i 's are distinct Fermat Primes.

Outline of Proof

Let ζ_n be a primitive n -th root of unity. It is clear

- the regular n -gon is constructible $\leftrightarrow \zeta_n$ is constructible.

This is the criterion we used to say that the regular n -gon is constructible when we implemented it into Lean.

- $\mathbb{Q} \subset \mathbb{Q}(\zeta_n)$ is a Galois Field extension.
- If $\alpha \in \mathbb{C}$ is algebraic over $\mathbb{Q}$, and $\mathbb{Q} \subset L$ is the splitting field of the minimal polynomial of α over $\mathbb{Q}$. Then α is constructible if and only if $[L : \mathbb{Q}]$ is a power of 2.

This implies that ζ_n is constructible if and only if $[\mathbb{Q}(\zeta_n) : \mathbb{Q}]$ is a power of 2.

Outline of Proof

Continuing the proof,

- $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$, where φ is the Euler Totient function.
- $\varphi(n)$ is a power of 2 if and only if

$$n = 2^s p_1 \cdots p_r$$

where $s \geq 0$ is an integer and the p_i 's are distinct Fermat Primes.

This chain of if and only if statements proves the theorem.

Summary of Steps to Formalize Gauss-Wantzel

1. $\mathbb{Q} \subset \mathbb{Q}(\zeta_n)$ is Galois: already in Mathlib
2. Theorem 3 (splitting field degree characterization of constructible numbers): not in Mathlib or Ludwig's library, although Ludwig did formalize Theorem 2, which is the main tool in proving Theorem 3
3. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$: mostly in Mathlib
4. $\varphi(n) = 2^m$ implies $n = 2^k s_1 \cdots s_\ell$ where s_i are distinct Fermat primes: Fermat numbers were very recently added to Mathlib, but the parts that have been formalized are not very helpful

General Issues

- Installing Ludwig's library: difficult due to mediocre lake documentation and having to rewrite parts of code to work with an updated version of Mathlib
- Mathlib updates: needed to access theorems from version of Mathlib newer than one used in the project GitHub, so copied missing theorems to our code when possible instead of updating and recompiling Mathlib

Specific Issues: Step 2

- Requires the following lemma: Given a group G such that $|G| = 2^n$, then there exists a subgroup tower $G_n = 1 \subset G_{n-1} \subset \cdots \subset G_1 \subset G_0 = G$ where G_{i+1} is normal in G_i and of index 2.
- This was formalized by a function $\text{Fin } (n+1) \rightarrow \text{Subgroup } G$
- $\text{Fin } (n+1)$ is a type with elements $0, 1, \dots, n$, and addition works via modular arithmetic
- The use of Fin made things difficult, as we had to cast between $\mathbb{N}$, $\text{Fin } (n+1)$, $\text{Fin } (n+1+1)$, and $\text{Fin } (n+2)$ and do addition operations - resulted in simple addition statements needing to be proved in frustrating way

Specific Issues: Step 2

- Additionally, given groups $H \subset K \subset G$, the distinction between H, K being viewed as instances of Subgroup G versus H as an instance of type subgroup K (where K is coerced to a type) also caused issues.
- Finally, proof requires the subgroups of index 2 are normal, which is not formalized in Mathlib yet surprisingly
- This final lemma also requires working with cosets directly, which can be very clunky in Lean

Specific Problems: Step 4

Deciding how to formalize

$$n = 2^s p_1 \cdots p_r$$

where $s \geq 0$ is an integer and the p_i 's are distinct Fermat Primes in a usable way turned out to be harder than expected.

- There are many ways of extracting the prime factors of a number n , with different lemmas built on these different ways.
- There was no definition of a Fermat Prime yet in Lean, this had to be formulated in a way that agreed with how we extract prime factors.

References

- [Cox12] David A Cox. *Galois theory*. John Wiley & Sons, 2012.
- [Mon24] Ludwig Monnerjahn. *Formalisation of Constructible Numbers*. <https://github.com/Louis-Le-Grand/Formalisation-of-constructable-numbers/tree/main>. 2024.

Fim da apresentação!